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Exploring cognitive diversity in a sample of children from Unsatisfied Basic Needs homes

Federico Giovannetti^a, Marcos Luis Pietto^{a,b}, Sebastián Javier Lipina^{a,c},
Juan Esteban Kamienkowski^d and María Soledad Segretin^{a,c}

^aUnidad de Neurobiología Aplicada (UNA, CEMIC-CONICET), Buenos Aires, Argentina; ^bUniversidad Argentina de la Empresa (UADE), Buenos Aires, Argentina; ^cUnidad Académica Instituto de Investigaciones IUC-CEMIC, Buenos Aires, Argentina; ^dLaboratorio de Inteligencia Artificial Aplicada (LIAA, FCEyN- CONICET), Buenos Aires, Argentina

ABSTRACT

Child diversity has become an increasingly relevant topic in developmental science with significant methodological and applied implications. Clustering algorithms, a type of machine learning approach, offer accessible methods for exploring such diversity by identifying the latent data structure, which allows for the extraction of groups or profiles of subjects. Following a person-oriented approach, this study explores the cognitive profiles of a sample of 105 children coming from Unsatisfied Basic Needs homes in Argentina. An ensemble clustering method was applied to performance data from tests assessing attention, inhibitory control, working memory, planning, and fluid reasoning. Results delivered three specific clusters with statistically significant differences in cognitive accuracy and reaction time variables. Specifically, we found a “working memory low-performance group,” a “generalized high-performance group,” and a “generalized low-performance group.” The individual characteristics of these clusters may potentially guide differentiated strategies for future interventions. In addition, the incorporation of diverse performance variables – such as accuracy and response times – can help uncover individual strengths and weaknesses that may inform the design of such interventions. This approach contributes to fostering more inclusive and diverse perspectives in developmental science, moving away from deficit-based interpretations of low or atypical performance and instead recognizing them as expressions of the potential diversity of developmental pathways.

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The study of diversity in developmental science

At present, developmental science shows a growing interest in exploring development from a more diverse and inclusive perspective (Nketia et al., 2021; Noble et al., 2021; Syed et al., 2018). This variability has been framed as individual differences, heterogeneity, diversity, person-centered or person-oriented approaches, amongst others. In that sense, and beyond the eventual commonalities inherent to the species, each person or group

might show unique characteristics that differentiate them from others and should be studied not necessarily as outliers or as indicators of a deficit, but as the result of the idiographic or individual laws of development (Lerner, 2018; Von Eye et al., 2015). From a Relational Developmental Systems (RDS) perspective, this poses diversity as a result of inter- and intra-individual change and plasticity across development, making it a key focus for interventions aimed at promoting child development (Bornstein, 2019; Lerner, 2007).

According to Von Eye et al. (2015), developmental science provides distinct approaches to studying individual development. This implies a continuum that ranges from differential psychology, which explores the individual differences of behavior and the contribution of environmental factors to such differences in determined populations; to idiographic research, which proposes individual-level analysis instead of aggregate-level analysis of psychological processes (e.g., Molenaar, 2004).

Amid these two perspectives, person-oriented research emerges as a relevant and feasible alternative for studying development (Bergman & Lundh, 2015; Von Eye et al., 2015). On one hand, it shares conceptual roots with idiographic research, as it rarely assumes that study samples originate from a single homogeneous population. On the other hand, it also acknowledges that some children may exhibit similar developmental or performance patterns, allowing researchers to identify specific populations or sub-groups within a study sample. This supports aggregate-level estimates without losing the individual-level information.

Given the relevance of diversity as a defining element of developmental processes, studying it requires methodological strategies aligned with theoretical frameworks that emphasize individual variability. In this context, the person-oriented approach not only offers a conceptual lens to understand developmental differences but, also, calls for analytical tools capable of capturing the complexity of individual profiles within heterogeneous samples. Researchers increasingly use methods that identify meaningful patterns within populations, treating variability as informative rather than as statistical noise. These approaches preserve the complexity of development and align individual-level data with theoretical models.

Exploration of diversity through clustering approaches

The last decades' developments in machine learning and data science have led to accessible and straightforward methods to analyze such patterns and sample diversity. Specifically, cluster analysis allows group extraction by identifying the latent structure of the data. According to Bergman et al. (2017), these methods are in line with the person-oriented perspective as they allow researchers to find classification patterns or profiles that take the holistic relations between the different characteristics (i.e., variables) at the individual level, and summarize the data structure without a priori assumptions about the sample.

Cluster analysis aims to group data points (e.g., subjects) according to their similarity, where every group contains at least one data point, and each data point belongs to only one group (Gao et al., 2023; Giordani et al., 2020). Cluster methods can be classified into two main approaches: *supervised clustering*, where groups are known a priori and the research objectives include predicting the cluster membership of new data; and

unsupervised clustering, in which the researcher has no a priori information about the number of optimal groups (Giordani et al., 2020).

Several unsupervised clustering methods are present in the current literature. For instance, they could be classified into hierarchical and nonhierarchical methods. Hierarchical procedures do not directly give a series of clusters but, instead, construct a tree of clusters or partitions that allow dividing bigger clusters into smaller ones, or agglomerating small clusters into bigger ones. Limitations of these methods include high computational costs with large datasets, low predictability for new datasets, and low stability when new data is added (Gao et al., 2023).

Nonhierarchical methods aim to discover the best solution given the number of desired clusters, which must be decided before running the analyses. These methods include density-based methods (mostly used for spatial data), Gaussian Mixture Models (usually implemented in complex data with high sample sizes), and center-based methods, among others. Center-based clustering methods, like k-means or partitioning around medoids (PAM) are the most implemented clustering methods probably because of their computational speed and easy interpretation. These methods imply selecting initial centers or centroids equal to the number of desired clusters and finding mutually exclusive groups with spherical shapes based on the distance to those centers. The most straightforward k-means algorithm (Ball & Hall, 1965; Macqueen, 1967) finds the optimal solution by first randomly selecting several cluster centers and then estimating and iterating through new centroids until it finds the groups with lower within-cluster distances between subjects (Gao et al., 2023; Giordani et al., 2020). Some limitations of k-means models include being sensitive to outliers and being unable to identify non-spherical shapes in data (Gao et al., 2023).

Given the different advantages and shortcomings of the aforementioned clustering algorithms, several methods have been developed to minimize such limitations. Among those, clustering ensemble methods compare and aggregate results from different models by first running separate models, and then pooling the results via a consensus function (Gao et al., 2023). Clustering ensembles are a promising method in the search for robustness and stability in clustering results. However, these methods have not been extensively applied in developmental science.

Clustering models are widely used in psychology, neuropsychology, and psychiatry (Astle et al., 2022; Clatworthy et al., 2005; Morris et al., 1981), generating relevant evidence about individual characteristics at the cognitive, neural, and environmental levels (Astle et al., 2018, 2022; Bathelt et al., 2018; Reid & Ready, 2022). However, some deficit-based statements are still being held in the interpretation of such results.

For instance, Yokota et al. (2015) grouped children according to different patterns of cognitive performance and brain structure. They administered the WISC III to typically developed Japanese children from 5 to 16 years old and implemented unsupervised k-means clustering models. Results delivered a six-cluster solution with 3 groups showing an average performance and three presenting a lower performance than the sample mean. These profiles were associated with distinct levels of gray matter volume in the right middle temporal gyrus.

In another study, diverse neuropsychological tasks were implemented in a sample of typically developed Brazilian children and adolescents from 7 to 14 years old (Gabrig et al., 2018). Both hierarchical and nonhierarchical clustering models were implemented,

giving a four-group solution with different cognitive performances. One group included the older children, who in turn showed higher performance values in organizational abilities, while the group including the younger children showed the lowest attentional performance. A study by Astle et al. (2018) applied Self-Organized Maps (SOM) and k-means clustering models to a sample of children aged 5 to 17 years old. Results delivered four distinct profiles with significant differences in the cognitive, academic, and neural levels. Specifically, they found three “deficit” profiles that included children showing lower-than-expected scores in cognitive, language, or math measures. A fourth group showed “age-appropriate” scores in all cognitive and academic measures.

Gabrig et al. (2018) as well as Astle et al. (2018) implemented relevant classification models, including a variety of measures that allowed the construction of complex and informative clusters. However, in both cases, results were framed on the relation between variables rather than the specific characteristics of children in the sample. In addition, conclusions still held a bias toward a linear explanation of results in the sense of higher and lower performances.

On the contrary, Younger et al. (2024) applied a person-centered approach and showed non-linear groupings in a sample of 7- to 15-year-old children that were classified according to their performances in the Adaptive Cognitive Evaluation Classroom (ACE-C) battery. The authors found 6 unique profiles that showed associations with math and reading abilities. As expected, higher performance values in all Executive Functions (EFs) implied better academic achievement. However, the relationship was not linear and, in some cases, profiles with higher performance in one process (e.g., working memory) but average performance in the others also demonstrated good academic achievements. For the authors, clustering methods may be suitable for exploring the strengths and weaknesses in each child and informing the design of interventions aimed at promoting cognitive development.

While the mentioned studies offer relevant insights into the use of clustering methods to identify cognitive diversity in different samples of children, they do not consider factors that have been previously associated with cognitive performance, such as temperamental traits (e.g., Rothbart et al., 2006) or socioeconomic status (Johnson et al., 2016; Noble & Giebler, 2020), amongst others. In the case of socioeconomic status, previous evidence showed that children from low-SES homes may show lower performance values in cognitive processing (e.g., Noble & Giebler, 2020). In the local context, such associations were particularly observed in tasks demanding cognitive self-regulation and executive functions such as attention, inhibitory control, working memory, cognitive flexibility, planning, and fluid reasoning, among others (Ison et al., 2015; Lipina et al., 2013).

Incorporating cluster analysis methods in the study of child development in low-SES samples could help improve the efficacy of cognitive intervention approaches in these populations. This could be highly important given that evidence shows that not all children participating in these kinds of interventions benefited equally from the activities (Katz & Shah, 2017). This raises the need for a better understanding of the different strengths and weaknesses of children in order to develop more personalized and adaptive interventions (Chaku et al., 2022; Plass & Pawar, 2020).

In spite of this, few studies involved clustering of cognitive profiles within low-SES samples. For example, Reid and Ready (2022) generated different cluster solutions based

on the sociodemographic characteristics of the children, and after that, they analyzed their cognitive performances. Results showed that all groups containing low-SES children exhibited lower cognitive performances when they entered kindergarten, but that Hispanic dual-language children with immigrant parents showed greater gains during development, showing different trajectories than other low-SES children.

In another study, Treviño et al. (2021) recruited typically developing preschool children from Mexico, Colombia, and Guatemala who completed a neuropsychological battery. The cluster analysis divided the sample into two distinct groups: a high-performance group with higher parental levels of education and a low-performance group with lower parental levels of education. This study is of great importance since it implements cluster analysis in a socioeconomically diverse sample. However, it fails to discover distinct performance profiles in children within high and low socioeconomic groups.

In light of the limitations of previous research, this study aims to explore the cognitive profiles of a sample of children from Unsatisfied Basic Needs homes (UBN, a low-socioeconomic status criterion used in Latin America) in Argentina. To our knowledge, evidence of clustering methods, including these kinds of populations, is scarce and limited. Following a person-oriented approach, we conducted clustering analyses to identify and describe differences at cognitive, temperamental, and contextual levels. Finally, the implications of these distinct patterns for promoting cognitive development are discussed.

Given the lack of this specific kind of evidence in literature, there are no *a priori* hypotheses on the expected cluster result. However, based on more broad literature on cognitive development, it would be reasonable that the different cluster profiles align with common accounts on executive functions' development (e.g., Bailey & Jones, 2019; Diamond, 2020; Garon et al., 2008). These, state that more simple processes such as attention, inhibitory control, and working memory; may develop earlier or support the development of more complex and integrative processes such as planning and reasoning. In that sense, we would expect that profiles of children with lower performance in simpler processes may also show lower values in more complex tasks. However, we could find children with moderate to high performance in tasks demanding simpler processes that do not necessarily transfer to higher performance in complex tasks. These developmentally-driven differences could be relevant targets for future intervention approaches.

Materials and methods

Participants

For this study, a non-probabilistic convenience sampling method was implemented. In agreement with the Ministry of Education of the City of Buenos Aires, a target population from UBN homes was selected. All participants attended a public kindergarten and had no history of developmental or neurological disorders. The original sample consisted of 117 children. Twelve children were excluded from the analyses due to low performance and/or missing data (more details in the *Data preprocessing* section). The removal of incomplete cases is essential for the clustering methods implemented later. The final

complete sample consisted of 105 participants (50 girls) with a mean age of 5.33 ($SD = 0.329$) years. Table 1 contains the final sample's sociodemographic characteristics.

All the implemented procedures complied with national and international research norms and were reviewed and approved by the IRB (CEMIC, Protocols N°682, 961). Parents or legal caregivers attended an informative talk about the characteristics of the study and gave their written consent for their children to participate in this research.

Procedure

All activities were performed in a 3×3 m ad hoc testing room that was only used by the research team. A maximum of three children at a time were invited to leave their classrooms and participate in the cognitive assessments. Each participant sat at a 1.30×0.5 m table separated by a wooden panel to avoid distractions. All activities were completed on a tablet (Samsung Galaxy Tab E with a 9.6" screen) placed at about 30 cm from the child, and connected to headphones to improve the auditory stimulus's reception and reduce eventual auditory interference. All participants had access to mobile tactile devices at home, as reported by their parents.

Children were accompanied by a research assistant who was blind to the goals and hypotheses of the study and was previously trained for task assessment. To mitigate the effects of fatigue and waning motivation, research assistants monitored participants after each task through direct inquiry and behavioral observation. Children were explicitly offered breaks if they expressed or exhibited signs of tiredness or disengagement.

Individual, temperament, and contextual variables

Information about individual and contextual characteristics of each child was gathered by administering a questionnaire aimed at obtaining information on SES and family living conditions (Lipina et al., 2013). First, to identify UBN homes, we applied a Socioeconomic Scale (Lipina et al., 2013). Following the criteria of the National Institute of Statistics and Censuses, a household was classified as having UBN if it

Table 1. Sample information.

Variables	
n	106
girls	51
Age	5.33 (0.33)
Parent education level ^a	6.97 (2.62)
Parent occupation level ^a	3.92 (1.69)
Housing ^b	10.37 (1.73)
Overcrowding ^c	7.59 (1.99)
SES Score	28.85 (4.66)
UBN	95.88%

^aHighest educational and occupational levels reached by parents (range 0 to 12 points).

^bScale range: 3–12 points, with higher scores for better housing conditions.

^cScale range: 0–9 points, with higher scores for better conditions.

presented at least one of the following: (1) inadequate housing; (2) absence of a waste disposal system; (3) overcrowding (more than three people per room); (4) school-aged children not attending any educational institution; or (5) a head of household with incomplete secondary education and more than four dependents.

Second, a total socioeconomic score was calculated based on parental education and occupation levels, overcrowding, housing conditions, and sanitation. Higher scores indicate better socioeconomic conditions, even within a UBN household.

In addition, temperament was assessed via the short form of the Child Behavior Questionnaire (CBQ; Putnam & Rothbart, 2006) which was administered to parents. This version has been previously validated for children between 3 and 7 years old and consists of 36 items referring to the responder's perceptions of the everyday behavior of a child. It taps three children's temperamental dimensions: *surgency* (distinguished by high impulsivity, activity, and low levels of shyness); *negative affect* (related to high levels of sadness, fear, anger and frustration), and *effortful control* (associated with high levels of inhibitory and attentional control). For this study, the reliability of each dimension was assessed via Cronbach's Alpha (*Surgency* = .71; *Negative affect* = .65; *Effortful control* = .59).

The individual, contextual, and temperamental variables that will be analyzed in this study include: age, sex, SES score, surgency, negative affect, and effortful control.

Cognitive assessments

The implemented assessment aimed at covering a broad set of EFs. These abilities ranged from simple processes such as attention, inhibitory control, and working memory to more complex and integrative processes such as planning and reasoning (Bailey & Jones, 2019; Diamond, 2020; Garon et al., 2008). These processes have been previously studied in 5-year-old children from low-SES populations (e.g., Hackman et al., 2015; Lipina et al., 2013; St. John et al., 2019) and in studies with cluster analysis approaches (e.g., Treviño et al., 2021; Yokota et al., 2015). As mentioned, the analyzed data is part of a larger project including the assessment of more tasks (Giovannetti et al., 2020; Giovannetti et al., 2022). Tasks included in this study were:

Attentional Networks Test for children (Child ANT): aimed to assess different aspects of attentional processes and executive control (Rueda et al., 2004). The current version consists of a screen with two buttons on the sides and a fixation cross in the center. Stimulus consisted of a row of 5 figures (fish, mouse, or bird) that appeared for 1700 ms, and the child had to press the button on the side that the middle figure was pointing to. Along three blocks of 96 trials, this figure would be randomly pointing to the same side as the rest (congruent trials) or to the opposite side (incongruent trials). Before the stimulus, some cues were presented for 150 ms as a white oval figure appearing in different parts of the screen. Reaction times (RTs) in this task allow the calculation of different scores related to distinct attentional networks/processes. The alerting network refers to the sensitivity to stimulus and vigilance; the orienting network refers to the ability to select relevant information; and the executive network implies monitoring and conflict resolution mechanisms of attention. For more information on how these scores are computed, refer to Rueda et al. (2004). The task included a practice block of 8 trials that could

be passed if the child had 5 correct trials; otherwise, children repeated the practice block once more. This task included several kinds of variables. On one hand, it included: the amount of completed trials by each child, the amount and percentage of omitted trials (i.e., trials without response), the amount of impulsive trials (i.e., trials with RT lower than 250 ms), amount and percentage of correct trials. All these variables were available for all trials and also disaggregated for congruent and incongruent trials. On the other hand, the task included RT variables, which contained mean and median RT for correct and incorrect trials. These RT were given for all trials, and also detailed for congruent and incongruent trials. Finally, the index values for the alerting network (RT for no-cue trials minus RT for double-cue trials), the orienting network (RT for central-cue trials minus RT for spatial-cue trials), and the executive network (RT for incongruent trials minus RT for congruent trials) were calculated. In sum, the ANT task delivered a total of 34 variables.

The direct CORSI Blocks task (Pickering, 2001): aimed to study spatial working memory (WM). The test consisted of a rectangular frame with nine blocks with a star inside that would light up in an established order. Children had to remember the exact sequence and press the buttons in the same order. Every four trials, the number of blocks increased progressively from one to eight blocks. The task stopped after three consecutive incorrect trials. This task included several variables of interest: the amount of completed trials, the amount and percentage of correct trials, the mean and median execution times for correct and incorrect trials, and the maximum achieved level (last level with more than 50% of correct trials). In sum, the CORSI task included a total of 8 variables.

The Tower of London (TOL) task: intended to measure planning and problem-solving abilities (Shallice, 1982). The current version presented three holes with descending depths. The first hole allowed only three balls inside it, the second two balls, and the third only one ball. Three balls were sorted into the holes, and children were told to move them to reach a required configuration. To accomplish the objective, children were allowed to move only one ball at a time, and only the top ball in each hole. The required number of moves increased with trials. The task stopped after five consecutive incorrect trials. For more details, see Giovannetti et al. (2022). This task included several variables of interest: the amount of completed trials, the amount and percentage of correct trials, the mean and median planing time (the time from the start of the trial until the first ball was placed in any hole) for correct and incorrect trials, and the maximum achieved level (last level with more than 50% of correct trials). In sum, the TOL task included a total of 8 variables.

Matrices task (KBIT): This test is part of the Kaufman Brief Intelligence Test (Kaufman & Kaufman, 1983) and was designed to measure fluid (non-verbal) reasoning. This was the only task administered manually with a printed workbook. Each trial consisted of non-verbal analogy problems with visual stimuli that were either figurative (people or objects) or abstract (such as geometrical shapes). The task required participants to understand the relation between the stimuli and point to the correct answer by selecting it among the incorrect options. In all cases, the task ended after a block of trials was entirely incorrect (five trials). The variables of interest were the amount of completed trials, the amount and percentage of correct trials, and the amount of incorrect trials (errors).

Statistical procedures

The statistical pipeline was inspired by Gao et al. (2023) guidelines for implementing cluster analyses in mental health research. The process involves data pre-processing, descriptive and pre-clustering analyses, selection of the optimal number of clusters, implementation of the clustering models, selection of a final solution, exploration of results, and external validation. All analyses code and supplementary materials are available at https://github.com/FedeGiovannetti/Giovannetti_et_al_cognitive_diversity_clustering and at <https://osf.io/97ksu>.

Data pre-processing

The data were pre-processed to prepare them for the subsequent analyses. First, some participants were excluded from the study due to not completing the task or showing very low performance, which suggested they might have had difficulties understanding the tasks. In particular, 7 children were removed due to not finishing ANT. Two participants were removed because they did not meet the cutoff criteria in CORSI (i.e., failing the first 3 trials). Four participants were excluded because they did not meet the cutoff criteria in TOL (i.e., failing the first five trials), while two were removed because they abandoned the task. Finally, two participants were excluded because they did not pass the first block of trials in KBIT, while another two abandoned the task.

After that, a selection of problematic variables was implemented by identifying those with a high amount of missing values and/or high intra-task correlation coefficients. Variables with more than 10% of missing observations were completely removed. After that, correlation analyses were conducted between all cognitive variables inside each task, and statistically significant associations ($p < .05$) and Person's r higher than .7 were identified. Given the extension of the correlation matrices, they were not included in the present manuscript but are available in the online supplemental material. A final number of 15 variables was selected for the main analyses. Included variables from ANT were: the amount of correct trials for congruent and incongruent trials, the amount of omitted trials for congruent and incongruent trials, the median RT for incongruent and incongruent trials, the alertness network, the orienting network, and the executive network. Included variables from CORSI were: amount of correct trials, the median of execution times for correct trials, and the median of execution times for incorrect trials. Included variables from TOL were: the amount of correct trials, and the planning times for correct trials. From KBIT, the only variable included was the amount of correct trials. For clarity, in the following, all variables referring to the amount of omitted or correct trials will be named as *accuracy variables*, while RT and other time variables will be named as *time variables*.

Additionally, a final removal of missing data was performed by excluding all cases with missing values in any of the selected variables. Imputation of those cases was not feasible, as in some instances the missing values corresponded to children who did not register any data for a specific task. In other cases, the cause of the missing values could not be inferred. A final dataset of 105 participants and 15 variables was formed. Finally, all variables were normalized to z values. Given that two ANT variables (omitted congruent trials and omitted incongruent trials) were inversely correlated with accuracy, they were inverted to facilitate the interpretation of results.

Pre-clustering analyses

First, a range of potential optimal number of clusters (k) was selected. According to literature guidelines (e.g., Gao et al., 2023), several values of k were tested from 2 to $\sqrt{\text{samplesize}}$. The classical elbow and silhouette method was first used to narrow the options to only three alternatives of k via the k -means method. According to Gao et al. (2023), the results of the pre-clustering tests should only be used as a guideline and not for hypothesis testing.

Second, several cluster ensemble methods with different algorithms were implemented via the diceR package (Chiu & Talhouk, 2025) in order to define both the value of k and the most suitable algorithms for the final ensemble methods. Namely, k -means, Partition around Medoids (PAM), Diana, and hierarchical cluster analyses were implemented with Euclidean and Manhattan distances. In the case of hierarchical clustering, distinct methods were implemented too (average, complete, ward.d, ward.d2).

Both the number of k and the included methods were decided via visual inspections of the consensus matrix heatmaps constructed via iterations of 100 clustering results for each algorithm, with 80% of randomly selected samples. All code and plots illustrating this process are available in the online supplementary materials.

Main analyses

After selecting the number of clusters (k) and algorithms, a final model was run, and the final clustering solution was obtained using the Link-based Cluster Ensemble (LCE) method. Cognitive performance, along with individual and contextual characteristics, was compared across clusters. Given that the assumptions for parametric methods were not met, clusters were compared using non-parametric approaches. Kruskal-Wallis tests were conducted, followed by pairwise Mann-Whitney U tests with Holm-Bonferroni corrections for post hoc comparisons. In the case of categorical variables, Fisher's exact test was performed.

Results

Pre-clustering analyses

The Elbow and Silhouette methods were applied to k -means models with 1 to 10 clusters. Visual inspection of the resulting indices narrowed the optimal number of clusters to a range between 2 and 4. After that, independent models for all selected algorithms, distances, and k were implemented for 100 iterations with random subsamples of 80% of the total sample. From there, consensus matrices were constructed and visualized via heatmaps. Visual inspection allowed the selection of three models: k -means, PAM with Euclidean distances, and PAM with Manhattan distances. In addition, a value of $k = 3$ was selected.

Main analyses

The final ensemble clustering model included the selected algorithms and k . Again, independent models for all selected algorithms, distances, and k were implemented for

100 iterations with random subsamples of 80% of the total sample for each algorithm. The final solution, constructed using the LCE criteria, delivered three distinct groups. Cluster 1 ($n = 39$) was labeled as a “working memory low-performance group”; Cluster 2 ($n = 23$) was designated as a “generalized high-performance group”; and Cluster 3 ($n = 43$) was a “generalized low-performance group.”

In order to understand the differences between each group, descriptive and inferential statistics were computed for each cognitive performance variable. Tables 2 and 3 present the results of the performance comparisons among the three groups. To simplify the results report, performance was grouped into accuracy and time variables (Figure 1).

Table 2. Descriptive and Kruskal-Wallis analyses for clustering result.

Task	Feature	Clusters	n	Mean	SE	H	p
ANT	Correct congruent trials	Cluster 1	39	34.103	1.025	42.018	<.001
		Cluster 2	23	38.739	1.226		
		Cluster 3	43	25.209	1.199		
	Correct incongruent trials	Cluster 1	39	19.231	1.091	46.193	<.001
		Cluster 2	23	28.348	1.555		
		Cluster 3	43	13.279	0.736		
	Omitted congruent trials	Cluster 1	39	8.615	0.682	38.591	<.001
		Cluster 2	23	4.304	0.611		
		Cluster 3	43	13.326	1.052		
	Omitted incongruent trials	Cluster 1	39	16.231	0.742	44.932	<.001
		Cluster 2	23	7.435	0.558		
		Cluster 3	43	19.465	1.172		
	RT congruent trials	Cluster 1	39	1,167.077	17.280	15.456	<.001
		Cluster 2	23	1,022.109	29.280		
		Cluster 3	43	1,158.023	30.502		
	RT incongruent trials	Cluster 1	39	1,285.333	20.877	18.506	<.001
		Cluster 2	23	1,112.130	37.953		
		Cluster 3	43	1,069.453	40.119		
	Alerting network	Cluster 1	39	95.641	23.198	0.041	0.98
		Cluster 2	23	97.174	26.432		
		Cluster 3	43	107.605	25.233		
	Orienting network	Cluster 1	39	-6.269	20.328	0.215	0.898
		Cluster 2	23	1.674	22.108		
		Cluster 3	43	-9.384	23.247		
	Executive network	Cluster 1	39	118.256	17.204	34.080	<.001
		Cluster 2	23	90.022	21.565		
		Cluster 3	43	-88.570	26.702		
CORSI	Correct trials	Cluster 1	39	5.974	0.549	22.934	<.001
		Cluster 2	23	13.609	1.447		
		Cluster 3	43	6.837	0.626		
	Execution time correct trials	Cluster 1	39	10.978	0.294	25.378	<.001
		Cluster 2	23	13.487	0.466		
		Cluster 3	43	13.340	0.354		
TOL	Execution time incorrect trials	Cluster 1	39	14.833	0.345	22.803	<.001
		Cluster 2	23	17.422	0.680		
		Cluster 3	43	17.792	0.475		
	Correct trials	Cluster 1	39	10.897	0.579	1.861	0.394
		Cluster 2	23	11.217	0.790		
		Cluster 3	43	9.605	0.724		
KBIT	Planning time correct trials	Cluster 1	39	5.623	0.250	4.883	0.087
		Cluster 2	23	5.239	0.320		
		Cluster 3	43	6.166	0.284		
	Correct trials	Cluster 1	39	12.282	0.444	10.558	0.005
		Cluster 2	23	13.739	0.686		
		Cluster 3	43	11.302	0.431		

Note. Bolded values indicate statistically significant differences at $p < .05$.

Table 3. Mann-Whitney U analyses for clustering results.

.Task	Feature	Cluster 1 vs. Cluster 2			Cluster 1 vs. Cluster 3			Cluster 2 vs. Cluster 3		
		z	p	r	z	p	r	z	p	r
ANT	Correct congruent trials	2.863	0.004	0.364	4.815	<.001	0.532	5.558	<.001	0.684
	Correct incongruent trials	4.058	<.001	0.515	3.948	<.001	0.436	6.240	<.001	0.768
	Omitted congruent trials	4.155	<.001	0.528	3.417	0.001	0.377	5.594	<.001	0.689
	Omitted incongruent trials	6.184	<.001	0.785	1.846	0.066	0.204	5.724	<.001	0.705
	RT congruent trials	3.840	<.001	0.488	0.413	0.683	0.046	3.277	0.002	0.403
	RT incongruent trials	3.811	<.001	0.484	3.589	0.001	0.396	0.787	0.435	0.097
	Alerting network	0.007	1.000	0.001	0.172	1.000	0.019	0.168	1.000	0.021
	Orienting network	0.444	1.000	0.056	0.172	1.000	0.019	0.343	1.000	0.042
CORSI	Executive network	1.202	0.232	0.153	5.358	<.001	0.592	4.152	<.001	0.511
	Correct trials	4.633	<.001	0.588	0.642	0.524	0.071	3.942	<.001	0.485
	Execution time correct trials	3.985	<.001	0.506	4.489	<.001	0.496	0.532	0.602	0.065
TOL	Execution time incorrect trials	3.257	0.002	0.414	4.554	<.001	0.503	0.639	0.530	0.079
	Correct trials	0.535	0.712	0.068	0.928	0.712	0.102	1.270	0.621	0.156
KBIT	Planning time correct trials	1.115	0.296	0.142	1.453	0.296	0.160	1.998	0.137	0.246
	Correct trials	2.114	0.070	0.268	1.473	0.142	0.163	3.137	0.005	0.386

Note. Bolded values indicate statistically significant differences at $p < .05$.

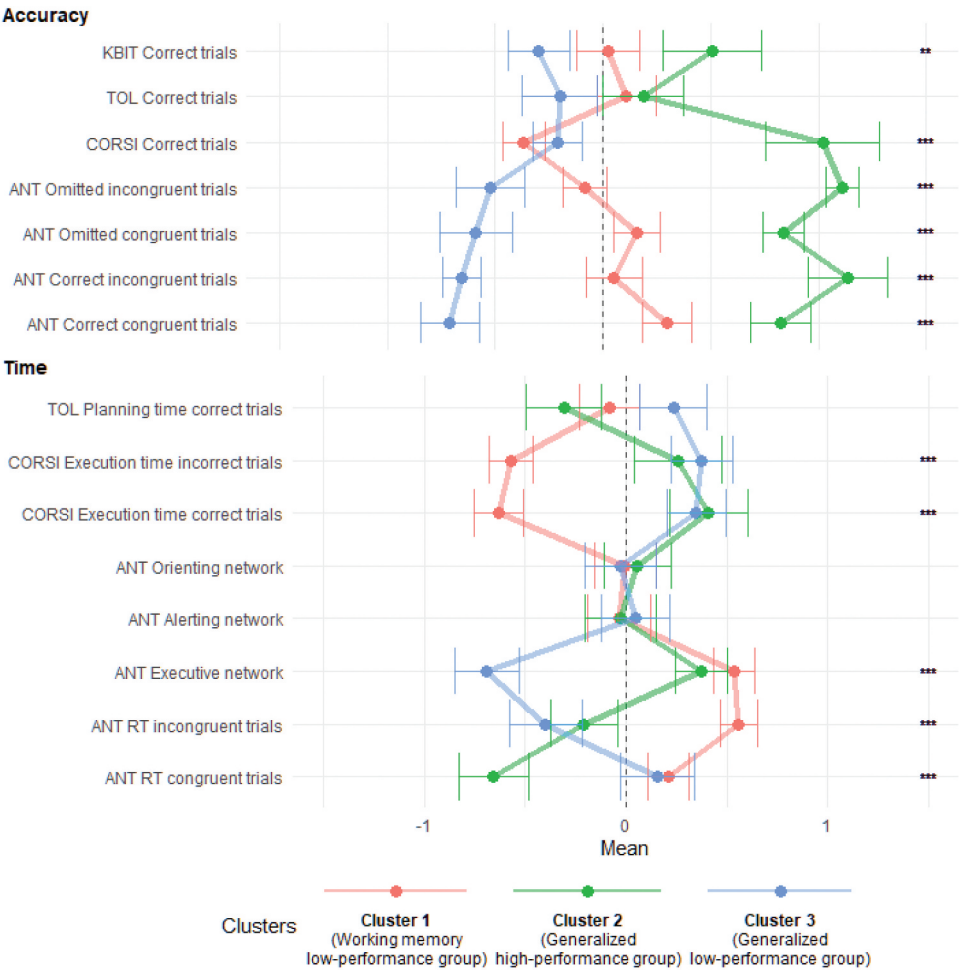


Figure 1. Cluster comparison of accuracy and times z-values.

Accuracy

In ANT, accuracy variables showed statistically significant differences among all groups (Table 2). Specifically, in both congruent and incongruent trials, the amount of correct trials (Table 3) in the *generalized high-performance group* was the highest, with z-values approximately 1 standard deviation above the sample mean. This was followed by the *working memory low-performance group*, which exhibited moderate z-values near zero across all accuracy variables; and the *generalized low-performance group*, which showed z-values of approximately 0.5 standard deviations below the sample mean.

The amount of omitted congruent trials also showed statistically significant differences among all groups (Table 2), which was confirmed by pairwise comparisons (Table 3). As seen in Figure 1, the *generalized high-performance group* showed the highest z-values (1 standard deviation above mean), followed by the *working memory low-performance group*, which showed z-values near zero; and the *generalized low-performance group*, with below-zero z-values. In the case of omitted incongruent trials, pairwise comparisons (Table 3) showed that the high-performance group again had the highest values, while the other two groups showed no differences.

In CORSI, the *generalized high-performance group* showed a higher amount of correct trials than the other groups (Table 3), placing itself 1 standard deviation above the mean. Those other groups did not show statistically significant differences, having below-zero z-values.

In TOL, no statistically significant differences were found (Table 2), with all groups showing near-zero z-values. Finally, in KBIT, accuracy was placed 0.5 standard deviations above the mean and showed statistically significant differences between the *generalized high-performance group* and the other clusters (Table 3), which were both placed near the sample mean.

Times

In ANT, RT for congruent trials showed statistically significant differences among groups. Specifically, pairwise comparisons (Table 3) showed that the *generalized high-performance group* had the lowest RT, with below-zero z-values, while the other two clusters showed similar around-zero values with no statistically significant differences between them. On the other hand, RT for incongruent trials also showed statistically significant differences among groups. Pairwise comparisons (Table 3) showed that the *working memory low-performance group* had the highest values, which were placed above zero, while the other two clusters showed no differences, exhibiting below-zero values. The executive network showed below-zero z-values for the *generalized low-performance group*, having statistically significant differences with the other two clusters, which in turn showed above-zero values without differences between them (Table 3). Alerting and orienting networks did not show statistically significant differences (Table 2).

In CORSI, the *working memory low-performance group* showed low execution times in correct and incorrect trials with statistically significant differences with the other two clusters (Table 3). That cluster showed lower-than-average values, and the remaining groups exhibit above-zero values. In the case of TOL, no statistically significant differences were found (Table 2).

Individual and contextual characteristics

In order to understand the individual, contextual, and temperamental characteristics of the clusters, further analyses, including age, sex, SES score, and temperament dimensions, were implemented. Results indicated that none of these variables showed statistically significant differences among Clusters.

Discussion

In the present study, we implemented cluster analysis for the exploration of cognitive profiles of a sample of children coming from UBN homes in Argentina. This was accomplished by implementing a clustering ensemble method that delivered a three-cluster result consisting of groups integrated by children with statistically significant differences between them in cognitive variables. No differences were found in temperamental and contextual variables. In the following paragraphs, these results are discussed and examined in the light of previous evidence on cluster exploration in children's cognitive performance. In addition, some considerations are made based on the possible implications of results for the design of future research in intervention studies.

As mentioned, the three resulting clusters showed different cognitive profiles. Cluster 2 may be labeled as a “generalized high-performance” group since it showed high accuracy scores in attention, memory, and reasoning. In ANT, these accuracy levels were accompanied by low and low-to-average RTs in congruent and incongruent trials. Specifically, the executive network showed high values ($M = 90.022$), implying that children in this group utilized more time to answer incongruent trials than congruent ones correctly. In CORSI, high scores were accompanied by high execution times in both correct and incorrect trials.

Cluster 1 could be labeled as “working memory low-performance group.” This group showed accuracy scores around the sample mean in almost all tasks, except for CORSI, where performances were lower than the mean. Times in this group were higher than the average in all trials of ANT, with higher values for incongruent trials, as expressed by the executive network's scores ($M = 118.258$). In CORSI, execution times were lower than average in both correct and incorrect trials.

Cluster 3 may be labeled as a “generalized low-performance group”. This group showed lower-than-average accuracy levels in attention, memory, and reasoning. In ANT, these accuracy levels were accompanied by lower-than-average RTs in incongruent trials, leading to high executive network scores and, presumably, less time dedicated to inhibiting the prepotent response. In CORSI, low accuracy was followed by higher execution times, which did not necessarily result in more time dedicated to solving the task correctly.

Even though almost all tasks included time variables, they should not be necessarily analyzed together since, in some cases, times express RTs, while in other tasks they relate to longer-range times that probably involve other cognitive processes (i.e., execution times were measured in CORSI, while planning times were included in TOL).

In ANT, clusters showed a clear trend. The lower accuracy levels shown in the *generalized low-performance group* were accompanied by lower RTs in incongruent

trials that might reflect these children's difficulties in engaging in inhibitory control. In the *working memory low-performance group*, the higher RTs might have allowed children to dedicate appropriate time to engage in inhibitory control and respond accurately at their own time; while lower RTs in the *generalized high-performance group* might show a development in the ability to engage in inhibitory control without taking so long, while still showing higher accuracy values. This goes in line with previous work (e.g., Mezzacappa, 2004; Rueda et al., 2004) showing that children might need to expend more time in incongruent trials to engage cognitive control accurately, while higher-performance and older children might maintain this tradeoff while lowering overall RTs. Although cluster assignment was not associated with age in our analyses, the observed performance patterns reflect individual differences in attentional control strategies that may have influenced task outcomes.

In CORSI, higher accuracy was accompanied by higher execution times in the *generalized high-performance group*. However, the relationship between accuracy and execution times might not be sufficiently clear since the other two clusters showed similar accuracy values but opposed execution times. While children in the *working memory low-performance group* showed lower times, children in the *generalized low-performance group* dedicated longer times without performance improvements. Until now, recall times in CORSI had not been extensively studied in the literature. However, there is strong evidence showing an association between accuracy and processing speed, by which, across development, higher accuracy is associated with decreasing times (Barrouillet & Camos, 2020). Nevertheless, different developmental patterns for distinct tasks and types of WM suggest that individual differences may arise. One possibility is that RTs shown in the clusters with lower performance may evidence distinct strategies adopted by children to cope with their performances (e.g., children in the *working memory low-performance group* may be eliciting rapid responses to prevent the decay in memory traces). Another possibility arises from the relation between WM and attention. In a relevant review on this subject, Astle and Scerif (2011) relate how variability in attentional control may lead to variability in spatial working memory. In that sense, our results show that the clusters with lower performances had statistically significant differences in attentional control variables which may in turn lead to differences in WM. Finally, given that CORSI trials did not have time limits, other factors such as mind-wandering and a decline in goal-oriented action may be present, especially in the *generalized low-performance group*, which showed higher RTs. In a relevant article, Domingue et al. (2022) analyzed the relationship between accuracy and speed in 29 datasets containing data from diverse cognitive tasks. They found an inconsistent relationship between these two aspects of cognitive performance. In particular, increases in time use did not always predict increases in accuracy values. In this sense, while the speed-accuracy tradeoff implies that manipulation of time pressures may affect accuracy, authors suggest that other factors may play by which longer times may be related to lower accuracy levels. In future research, the association between accuracy and execution times should be analyzed by the implementation of correlation or regression analyses that might uncover distinct inter- and intra-cluster dynamics between these two variables.

In TOL, neither accuracy nor planning times showed differences between clusters. This could mean that this task may not be appropriate for studying planning or problem

solving in children with the characteristics of our sample (i.e., SES, age). Alternatively, it could indicate that our clustering solution was more effective in capturing differences related to simpler, but not to more complex, cognitive processes, which would require the integration of several ones (Bailey & Jones, 2019; Diamond, 2020; Garon et al., 2008). Furthermore, it is important to address that differences are not always expected, as there may be relevant variables not included in the analysis that better account for individual variability. Even though statistically significant differences were not found, it should be noted that children in the *generalized low-performance group* showed similar time values to those obtained in CORSI, which might support the idea of longer execution times related to mind wandering.

In relation to previous literature, our results align with those reported by Gabrig et al. (2018), where clustering models with a large sample of typically developed Brazilian children were implemented. In this case, attentional tasks showed an association with low performance groups, while organizational tasks were related to older and high-performance clusters. This might allow us to understand why planning did not show differences between clusters in our data while attentional tasks did. Following an integrated view of executive functions and self-regulation (Bailey & Jones, 2019; Diamond, 2020; Garon et al., 2008), it makes sense that children with lower attention and WM performances also showed lower performances in more complex abilities such as planning and reasoning. In the same vein, higher performance in these tasks may be highly influenced by simpler performance processes such as attention and cognitive control. As previously mentioned, future studies aimed at identifying cognitive diversity should explore the relationships between different cognitive processes both within and between clusters. This could have important implications for designing intervention studies aimed at exploring the integrative nature of cognition, considering that potentially more basic processes (e.g., attention) may also support the development of more complex ones (e.g., planning).

Concerning temperamental variables, the absence of differences could be related to the low reliability values found in Cronbach's Alpha tests. In that sense, a recent study conducted with samples with similar characteristics (Hermida et al., 2025) failed to find the expected dimensions from the original Rothbart & Bates' model. This was found especially with parent report measures, which differed significantly from teachers' reports. In future studies, temperamental measures could include reports coming from different caregivers in order to improve reliability values.

Finally, it should be noted that the resulting clusters did not show associations with any of the contextual factors studied. Even though the association between cognitive development and socioeconomic status has long been established (Johnson et al., 2016; Noble & Giebler, 2020), evidence of this relationship in studies that implemented cluster analyses is still scarce. For instance, Reid and Ready (2022) implemented a model that included different sociodemographic characteristics and found that children in a Low-SES cluster elicited lower cognitive performances and showed different developmental trajectories. In another study, Treviño et al. (2021) implemented a cluster solution that divided the sample into a high-performance group with a higher parental level of education and a low-performance group with a lower parental level of education. One hypothetical explanation of our results could be related to our sample SES homogeneity since all children lived in the same

neighborhood and went to the same school. However, this makes our results more interesting since, even with such homogeneity levels, we were able to find relevant differences between clusters at the cognitive level. This implies that there might be some level of cognitive diversity that might be independent of variations within low SES values and could be addressed by intervention programs aimed at promoting child development. In addition, there could be other variables that were not measured that could have been associated with differences between clusters.

Limitations

Several limitations should be noted for the present study. First, not all possible clustering algorithms were explored, and several well-known ones were left aside. For instance, DBSCAN is a relevant density-based clustering algorithm that is very common in computer and data sciences. However, it was discarded from our analysis for two reasons: One reason relates to the fact that density-based clustering is still uncommon in the area. Another reason has to do with the exploratory nature of the current article. We decided to focus only on hierarchical and center-based algorithms, which are both distance-centered. This may have been a relevant limitation, as partition-based algorithms tend to produce cluster solutions that reflect normative divisions – separating performance into low, medium, and high groups. Nonetheless, the inclusion of time-related variables might have allowed for a more complex interpretation of the results. Future studies should explore the use of other types of algorithms, including the combination of density and partition-based algorithms.

Second, it should be noted that the generalizability of our approach to other samples is still pending, since we explored only a specific sample with 105 participants with specific demographic characteristics, ages, and assessment tasks. Future work should implement the same approach with several and sufficiently diverse samples in order to extend the implications of this procedure. In this sense, several questions arise for such explorations: Will other samples exhibit similar clustering patterns? What differences and similarities might emerge between the characteristics of the current groups and those identified in new samples? How can we extend our approach to ensure it remains sufficiently flexible to adapt to different datasets while preserving interpretability?

Third, our analyses focused on a single-time assessment of cognitive processes based on specific tasks. This may fail to address relevant developmental dynamics within and between subjects, clusters, and cognitive processes themselves. The consideration of a longitudinal perspective in future studies might facilitate the interpretation of contradictory results and better inform intervention programs.

Fourth, statistically significant differences emerged exclusively for the cognitive measures – the same variables used to define the clustering models. While this might be expected given that clustering algorithms partition data based on these specific features, it is not a guaranteed outcome. Because these methods optimize for differences across a multi-dimensional space rather than on an individual variable basis, it is possible for specific features to remain non-significant within the final solution. Indeed, several of our cognitive measures did not differ significantly across clusters, suggesting that at this developmental stage, distinct profiles do not necessarily entail global divergence across all cognitive processes.

Fifth, the lack of associations with contextual factors may be related to our sample not being sufficiently heterogeneous in terms of these kinds of variables. Indeed, all children lived in the same neighborhood and, in some cases, even on the same block. In addition, other aspects of relevance for acknowledging diversity in development such as cultural or ethnic variability were not addressed. Future studies should include more diverse datasets to better explore potential associations between cognitive profiles and living contexts.

Conclusions

In the current study, we examined the cognitive profiles of children living in UBN households in Argentina. Literature on cognitive development of children coming from low-SES homes states that these children may show lower performance values in cognitive tasks demanding self-regulatory and executive functions (Johnson et al., 2016; Noble & Giebler, 2020). However, there is a lack of evidence-based guidance regarding how to systematically study the specific strengths and weaknesses of children living in these contexts and how such diversity could inform personalized and adaptive intervention programs.

Given that our approach was grounded in unsupervised learning methods, we held no *a priori* hypotheses for the clustering results. However, from a broader perspective on the development of executive functions, we were open to finding some specific profiles. For instance, we expected children with lower performance values in simpler processes to also show lower values in more complex tasks. In addition, we expected to find that children with moderate to high performance values in simpler processes would not necessarily transfer that performance to more complex tasks. This would be in line with common accounts on executive functions' development (e.g., Bailey & Jones, 2019; Diamond, 2020; Garon et al., 2008) that state that simpler processes, such as attention, inhibitory control, and working memory, may develop earlier or support the development of more complex and integrative processes such as planning and reasoning.

In the next paragraphs, we will address how the clustering result may align (or not) with those predictions; and how they could inform cognitive intervention programs as well as other intervention contexts such as schools, clinical neuropsychology, and public policy. Three specific clusters were identified, potentially guiding differentiated strategies for future interventions.

In the *generalized low-performance group* children showed lower accuracy values in all cognitive tasks and processes. As mentioned earlier, it could be expected that children with lower performances in tasks such as the ANT, which taps simple attentional and inhibitory control, would also show lower performance values in tasks requiring working memory or fluid reasoning. In addition, this profile showed rather erratic time values, as children responded faster to the attention task and responded more slowly to the working memory task. This suggests that children might have problems approaching both types of tasks, and several intervention approaches may be followed. Regarding the simple vs. complex processes vein: educators, practitioners, or researchers working with this kind of profile may focus primarily on the simpler processes as it is expected that, for instance, an improvement in attentional control may support higher working memory and reasoning performances. Regarding the time-use profiles,

interventions for these children could focus on enhancing time management skills, which may involve reducing impulsivity when appropriate and sustaining attentional focus during tasks such as CORSI, which may have led to some mind-wandering. In summary, these children may benefit from interventions targeting simple but essential cognitive control abilities.

In the *working memory low-performance group*, children showed lower performance only in CORSI, which does not clearly align with the theoretical accounts mentioned earlier. From such perspectives, we expected children with lower WM performances to also show lower performances in planning or fluid reasoning. However, it could be stated that the kind of WM processing assessed in the CORSI task may not necessarily overlap with the kind of WM needed in the KBIT. In our WM task, children were told to remember the position of the stimuli, but the KBIT required children to reason about the content of the stimuli. In future studies, the assessment of these two processes may be better aligned. Regarding the practical implications of this specific cognitive profile, tailored interventions could focus on activities promoting WM. In addition, reduction of impulsivity could also be targeted since low execution times were common in this group. In addition, given that performance on all other tasks was close to the sample mean, interventions could leverage these children's attentional, planning, and reasoning abilities to support improvements in working memory performance.

Finally, the *generalized high-performance group* aligns with our predictions since higher performance in simple processes was accompanied with higher performance in complex ones. In this sense, it could be decided whether this specific group of children should be part of an intervention at all. It could be the case that children in this group may already be at ceiling levels. Another reason may be that the tasks implemented in our work were not hard enough for them. Another possibility could be to give these children the opportunity to help children in the other clusters during the intervention, as in learning-by-teaching approaches (e.g., Hermida et al., 2021).

In sum, in the present work, we showed the implementation of an unsupervised clustering method for exploring cognitive diversity in a sample of children from adversity contexts. In addition, we presented a possible way of theoretically interpreting such profiles and a way of translating those interpretations to practical advice for intervention. This could be of great value for researchers, educators, practitioners, and also for public policy since it may give a blueprint to systematically identify individuals with similar performance profiles within a study sample and uncover individual strengths and weaknesses that may inform the design of tailored interventions. This approach contributes to fostering more inclusive and diverse perspectives in developmental science, moving away from deficit-based interpretations of low or atypical performance and instead recognizing them as expressions of the potential diversity of developmental pathways.

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Author contributions

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ORCID

Federico Giovannetti  <http://orcid.org/0000-0003-2238-3674>

References

- Astle, D. E., Bathelt, J., Team, T. C., & Holmes, J. (2018). Remapping the cognitive and neural profiles of children who struggle at school. *Developmental Science*, 22(1), e12747. <https://doi.org/10.1111/desc.12747>
- Astle, D. E., Holmes, J., Kievit, R., & Gathercole, S. E. (2022). Annual research review: The transdiagnostic revolution in neurodevelopmental disorders. *Journal of Child Psychology and Psychiatry*, 63(4), 397–417. <https://doi.org/10.1111/jcpp.13481>
- Astle, D. E., & Scerif, G. (2011). Interactions between attention and visual short-term memory (VSTM): What can be learnt from individual and developmental differences? *Neuropsychologia, Attention and Short-Term Memory*, 49(6), 1435–1445. <https://doi.org/10.1016/j.neuropsychologia.2010.12.001>
- Bailey, R., & Jones, S. M. (2019). An integrated model of regulation for applied settings. *Clinical Child & Family Psychology Review*, 22(1), 2–23. <https://doi.org/10.1007/s10567-019-00288-y>
- Ball, G. H., & Hall, D. J. (1965). *ISODATA, a novel Method of Data analysis and pattern classification*. Stanford Research Institute.
- Barrouillet, P., & Camos, V. (2020). The time-based resource-sharing model of working memory. In P. Barrouillet & V. Camos (Eds.), *Working memory* (pp. 85–115). Oxford University Press. <https://doi.org/10.1093/oso/9780198842286.003.0004>
- Bathelt, J., Holmes, J., Astle, D. E., Holmes, J., Gathercole, S., Astle, D., Manly, T., Kievit, R., & Centre for Attention Learning and Memory. (2018). Data-driven subtyping of executive

- function-related behavioral problems in children. *The Journal of the American Academy of Child & Adolescent Psychiatry*, 57(4), 252–262.e4. <https://doi.org/10.1016/j.jaac.2018.01.014>
- Bergman, L. R., & Lundh, L.-G. (2015). Introduction: The person-oriented approach: Roots and roads to the future. *Journal for Person-Oriented Research*, 1(1–2), 1–6. <https://doi.org/10.17505/jpor.2015.01>
- Bergman, L. R., Vargha, A., & Kövi, Z. (2017). Revitalizing the typological approach: Some methods for finding types. *Journal for Person-Oriented Research*, 3(1), 49–62. <https://doi.org/10.17505/jpor.2017.04>
- Bornstein, M. H. (2019). Fostering optimal development and averting detrimental development: Prescriptions, proscriptions, and specificity. *Applied Developmental Science*, 23(4), 340–345. <https://doi.org/10.1080/10888691.2017.1421424>
- Chaku, N., Barry, K., Fowle, J., & Hoyt, L. T. (2022). Understanding patterns of heterogeneity in executive functioning during adolescence: Evidence from population-level data. *Developmental Science*, 25(6). <https://doi.org/10.1111/desc.13256>
- Chiu, D., & Talhouk, A. (2025). *dicER: Diverse cluster ensemble in R*. <https://github.com/AlineTalhouk/diceR/>
- Clatworthy, J., Buick, D., Hankins, M., Weinman, J., & Horne, R. (2005). The use and reporting of cluster analysis in health psychology: A review. *British Journal of Health Psychology*, 10(Pt 3), 329–358. <https://doi.org/10.1348/135910705X25697>
- Diamond, A. (2020). Executive functions. *Handbook of Clinical Neurology*, 173, 225–240. <https://doi.org/10.1016/B978-0-444-64150-2.00020-4>
- Domingue, B. W., Kanopka, K., Stenhaus, B., Sulik, M. J., Beverly, T., Brinkhuis, M., Circi, R., Faul, J., Liao, D., McCandliss, B., Obradović, J., Piech, C., Porter, T., Consortium, P. iLEAD, Soland, J., Weeks, J., Wise, S. L., & Yeatman, J. (2022). Speed-accuracy trade-off? Not so fast: Marginal changes in speed have inconsistent relationships with accuracy in real-world settings. *Journal of Educational and Behavioral Statistics*, 47(5), 576–602. <https://doi.org/10.3102/10769986221099906>
- Gabrig, I. A., Oliveira, R. M., & Charchat-Fichman, H. (2018). Subgroups in cognitive development: Distribution of Brazilian students aged 7–14 years. *Psychology and Neuroscience*, 11(2), 168–179. <https://doi.org/10.1037/pne0000112>
- Gao, C. X., Dwyer, D., Zhu, Y., Smith, C. L., Du, L., Filia, K. M., Bayer, J., Menssink, J. M., Wang, T., Bergmeir, C., Wood, S., & Cotton, S. M. (2023). An overview of clustering methods with guidelines for application in mental health research. *Psychiatry Research*, 327, 115265. <https://doi.org/10.1016/j.psychres.2023.115265>
- Garon, N., Bryson, S. E., & Smith, I. M. (2008). Executive function in preschoolers: A review using an integrative framework. *Psychological Bulletin*, 134(1), 31–60. <https://doi.org/10.1037/0033-2909.134.1.31>
- Giordani, P., Ferraro, M. B., & Martella, F. (2020). *An introduction to clustering with R* (Vol. 1). Springer Singapore. <https://doi.org/10.1007/978-981-13-0553-5>
- Giovannetti, F., Pietto, M. L., Segretin, M. S., & Lipina, S. J. (2022). Impact of an individualized and adaptive cognitive intervention on working memory, planning and fluid reasoning processing in preschoolers from poor homes. *Child Neuropsychology*, 28(5), 597–626. <https://doi.org/10.1080/09297049.2021.1998406>
- Giovannetti, F., Pietto, M. L., Segretin, M. S., & Lipina, S. J. (2020). Impact of an individualized cognitive training intervention in preschoolers from poor homes. *International Journal of Environmental Research and Public Health*, 17(8), 1–20. <https://doi.org/10.3390/ijerph17082912>
- Hackman, D. A., Gallop, R., Evans, G. W., & Farah, M. J. (2015). Socioeconomic status and executive function: Developmental trajectories and mediation. *Developmental Science*, 18(5), 686–702. <https://doi.org/10.1111/desc.12246>
- Hermida, M. J., Perez Santangelo, A., Calero, C. I., Goizueta, C., Espinosa, M., & Sigman, M. (2021). Learning-by-teaching approach improves dengue knowledge in children and parents. *The American Journal of Tropical Medicine and Hygiene*, 105(6), 1536–1543. <https://doi.org/10.4269/ajtmh.21-0253>

- Ison, M. S., Greco, C., Korzeniowski, C., & Morelato, G. (2015). Selective attention: A comparative study on Argentine students from different socioeconomic contexts. *Electronic Journal of Research in Educational Psychology*, 13(2), 343–368.
- Johnson, S. B., Riis, J. L., & Noble, K. G. (2016). State of the art review: Poverty and the developing brain. *Pediatrics*, 137(4), e20153075. <https://doi.org/10.1542/peds.2015-3075>
- Katz, B., & Shah, P. (2017). The role of child socioeconomic status in cognitive training outcomes. *Journal of Applied Developmental Psychology*, 53(July 2016), 139–150. <https://doi.org/10.1016/j.appdev.2017.10.003>
- Kaufman, A. S., & Kaufman, N. L. (1983). *Kaufman assessment battery for children*. John Wiley & Sons, Inc.
- Lerner, R. M. (2007). Developmental science, developmental systems, and contemporary theories of human development. In W. Damon & R. M. Lerner (Eds.), *Handbook of child psychology* (1st ed., pp. 1–17). Wiley. <https://doi.org/10.1002/9780470147658.chpsy0101>
- Lerner, R. M. (2018). *Concepts and theories of human development* (4th ed.). Routledge. <https://doi.org/10.4324/9780203581629>
- Lipina, S., Segretin, S., Hermida, J., Prats, L., Fracchia, C., Camelo, J. L., & Colombo, J. (2013). Linking childhood poverty and cognition: Environmental mediators of non-verbal executive control in an Argentine sample. *Developmental Science*, 16(5), 697–707. <https://doi.org/10.1111/desc.12080>
- Macqueen, J. (1967). Some methods for classification and analysis of multivariate observations. *Proceedings of the 5th Berkeley Symposium on Mathematical Statistics and Probability*.
- Mezzacappa, E. (2004). Alerting, orienting, and executive attention: Developmental properties and sociodemographic correlates in an epidemiological sample of young, urban children. *Child Development*, 75(5), 1373–1386. <https://doi.org/10.1111/j.1467-8624.2004.00746.x>
- Molenaar, P. C. M. (2004). A manifesto on psychology as idiographic science: Bringing the person back into scientific psychology, this time forever. *Measurement: Interdisciplinary Research & Perspective*, 2(4), 201–218. https://doi.org/10.1207/s1536/6359mea0204_1
- Morris, R., Blashfield, R., & Satz, P. (1981). Neuropsychology and cluster analysis: Potentials and problems. *Journal of Clinical Neuropsychology*, 3(1), 79–99. <https://doi.org/10.1080/01688638108403115>
- Nketia, J., Amso, D., & Brito, N. H. (2021). Towards a more inclusive and equitable developmental cognitive neuroscience. *Developmental Cognitive Neuroscience*, 52, 101014. <https://doi.org/10.1016/j.dcn.2021.101014>
- Noble, K. G., & Giebler, M. A. (2020). The neuroscience of socioeconomic inequality. *Current Opinion in Behavioral Sciences*, 36, 23–28. <https://doi.org/10.1016/j.cobeha.2020.05.007>
- Noble, K. G., Hart, E. R., & Sperber, J. F. (2021). Socioeconomic disparities and neuroplasticity: Moving toward adaptation, intersectionality, and inclusion. *The American Psychologist*, 76(9), 1486–1495. <https://doi.org/10.1037/amp0000934>
- Pickering, S. J. (2001). The development of visuo-spatial working memory. *Memory*, 9(4–6), 423–432. <https://doi.org/10.1080/09658210143000182>
- Plass, J. L., & Pawar, S. (2020). Adaptivity and personalization in games for learning. In J. L. Plass, R. E. Mayer, & B. D. Homer (Eds.), *Handbook of Game-Based Learning* (pp. 263–282). MIT Press.
- Putnam, S. P., & Rothbart, M. K. (2006). Development of short and very short forms of the children's behavior questionnaire. *Journal of Personality Assessment*, 87(1), 102–112. https://doi.org/10.1207/s15327752jpa8701_09
- Reid, J. L., & Ready, D. D. (2022). Heterogeneity in the development of executive function: Evidence from nationally representative data. *Early Education & Development*, 33(2), 243–267. <https://doi.org/10.1080/10409289.2020.1866949>
- Rothbart, M. K., Posner, M. I., & Kieras, J. (2006). Temperament, attention, and the development of self-regulation. In K. McCartney & D. Phillips (Eds.), *Blackwell handbook of early childhood development* (pp. 338–357). Blackwell Publishing Ltd. <https://doi.org/10.1002/9780470757703.ch17>

- Rueda, M. R., Fan, J., McCandliss, B. D., Halparin, J. D., Gruber, D. B., Lercari, L. P., & Posner, M. I. (2004). Development of attentional networks in childhood. *Neuropsychologia*, 42 (8), 1029–1040. <https://doi.org/10.1016/j.neuropsychologia.2003.12.012>
- Shallice, T. (1982). Specific impairments of planning. *Philosophical transactions of the Royal Society of London. B, Biological Sciences*, 298 (1089), 199–209. <https://doi.org/10.1098/rstb.1982.0082>
- St. John, A. M., Kibbe, M., & Tarullo, A. R. (2019). A systematic assessment of socioeconomic status and executive functioning in early childhood. *Journal of Experimental Child Psychology*, 178, 352–368. <https://doi.org/10.1016/j.jecp.2018.09.003>
- Syed, M., Santos, C., Yoo, H. C., & Juang, L. P. (2018). Invisibility of racial/ethnic minorities in developmental science: Implications for research and institutional practices. *The American Psychologist*, 73(6), 812–826. <https://doi.org/10.1037/amp0000294>
- Treviño, M., Beltrán-Navarro, B., León, R. M.-C. Y., & Matute, E. (2021). Clustering of neuropsychological traits of preschoolers. *Scientific Reports*, 11(1), 6533. <https://doi.org/10.1038/s41598-021-85891-2>
- Von Eye, A., Bergman, L. R., & Hsieh, C. (2015). Person-oriented methodological approaches. In R. M. Lerner (Ed.), *Handbook of child psychology and developmental science* (1st ed., pp. 1–53). Wiley. <https://doi.org/10.1002/9781118963418.childpsy121>
- Yokota, S., Takeuchi, H., Hashimoto, T., Hashizume, H., Asano, K., Asano, M., Sassa, Y., Taki, Y., & Kawashima, R. (2015). Individual differences in cognitive performance and brain structure in typically developing children. *Developmental Cognitive Neuroscience*, 14, 1–7. <https://doi.org/10.1016/j.dcn.2015.05.003>
- Younger, J. W., Schaerlaeken, S., Anguera, J. A., & Gazzaley, A. (2024). The whole is greater than the sum of its parts: Using cognitive profiles to predict academic achievement. *Trends in Neuroscience and Education*, 36, 100237. <https://doi.org/10.1016/j.tine.2024.100237>